

Network Troubleshooting

Learning Objectives

- Identify common network problems
- Use basic troubleshooting steps
- Understand when to seek technical help

Engage (5-7 minutes)

When the water supply stops in your house, you check the tap, then the pipe, then call the plumber. Network problems are similar—first check the simple things like cables and Wi-Fi, then move to bigger issues. Today we learn how to become our own network plumbers.

Explore (10-12 minutes)

Your teacher will simulate a network problem—disconnect a cable or turn off Wi-Fi. Try to find the problem using these steps: Is the cable plugged in? Is Wi-Fi turned on? Can you ping another computer? Write down what you find at each step.

Explain (10-12 minutes)

Common network problems include loose cables, disabled Wi-Fi, incorrect passwords, and router issues. Basic troubleshooting steps are: check physical connections, restart your computer and router, verify network settings, and test with another device. If the problem persists, contact your network administrator or ISP.

Elaborate (8-10 minutes)

Think of fixing a bicycle. First check if the tire is flat, then the chain, then the brakes—start simple before assuming the worst. Network troubleshooting follows the same logic: check the obvious first. If your computer cannot connect to Wi-Fi, what are the first three things you should check?

Evaluate (5-8 minutes)

Exit ticket: List four steps to troubleshoot a network connection problem. Then write one situation where you would need to call for technical help. If you finish early, practice the ping command to test a connection.